

Prometheus - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://prometheus.io/download/>

What Prometheus is

Prometheus collects metrics from targets, usually by scraping HTTP endpoints, and stores the values as time series with timestamps and labels.

Scraping and targets

The configuration file defines targets and collection frequency. Before interpreting charts, confirm that the target is UP and that the metric is actually being exposed and collected.

PromQL

PromQL is the query language used to select and aggregate time series. Start with simple queries and only then move to rates, aggregations and time windows.

QA and observability

During performance, integration or post-deploy validation, QA can correlate functional behavior with latency, errors, resource consumption and other metrics exposed by the system.

Alerts and context

Metrics can feed rules and alerts. A useful alert must represent an actionable condition; too many alerts without context create noise.

Attention point

Prometheus does not understand product quality by itself. Metrics need coherent names, labels and objectives; excessive cardinality can also create cost and complexity.